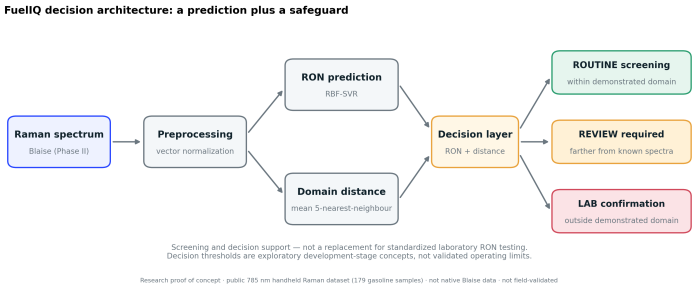
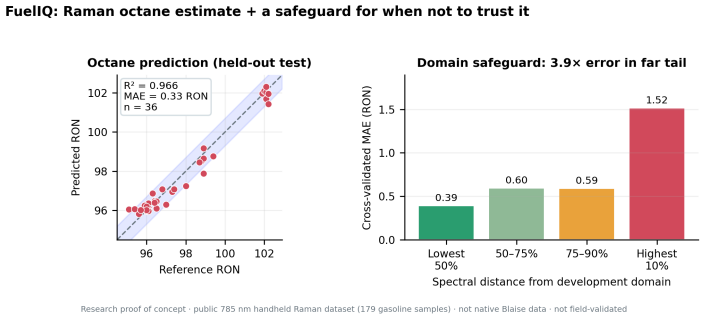
FuelIQ: Raman Gasoline Octane Screening with a Built-In Trust Check

Call to action: Octane checks today are either a lab sample sent away or a number nobody can tell whether to trust; FuelIQ puts a Raman-based octane estimate and an applicability-domain safeguard in the field, so users know when routine screening is enough and when to confirm in a lab.





1. PROBLEM STATEMENT

A gasoline's octane rating decides whether it protects an engine from knock, yet the standard Research Octane Number (RON) test needs a dedicated test engine in a laboratory, so quality checks are slow and sample-based. Field tools that return a number have a second problem: they give a confident-looking value even when the sample is unlike anything the model was built on, and the user cannot tell when not to trust it.

2. PROPOSED SOLUTION

FuelIQ pairs the Blaise pocket Raman spectrometer with a phone app. The user scans a fuel sample; the app normalizes the spectrum, estimates RON with a trained regression model, and in parallel measures how far the spectrum sits from the spectra the model was built on. The result shows the estimate and one of three statuses: Routine screening, Review required, or Laboratory confirmation. It is screening and decision support, and it keeps a model estimate clearly separate from a laboratory-certified measurement. Video: https://youtube.com/watch?v=6xwVU_PyTNk&ra;=m

3. UNIQUENESS AND INNOVATION

The novelty is not just predicting octane from Raman; it is shipping the prediction together with a safeguard. A nearest-neighbor spectral-distance check runs beside the model. In leakage-free 5-fold cross-validation on 143 samples, distance and error correlated at r = 0.48, and the farthest 10% of samples had 3.9x the error of the closest half (MAE 1.52 vs 0.39 RON). That is a useful warning signal, not a guarantee, and the decision thresholds are still exploratory.

4. IMPACT AND SCALABILITY

Intended users are fleet and depot quality staff, fuel distributors and retailers, and inspectors who need a fast first screen and a clear rule for when to send a sample to the lab. The pipeline is software-defined, so more fuels and properties can be added as paired Blaise spectra and lab references accumulate, and each validated scan can strengthen the shared model.

5. FEASIBILITY AND TECHNICAL APPROACH

Evidence so far (public FuelRamanSpectraHandheld dataset: 179 commercial gasoline samples, 1,901 Raman variables, 400-2300 cm^{-1}): vector normalization plus RBF support-vector regression, tuned on validation data only and frozen before the official 36-sample test, reached R^2 0.966, MAE 0.334 RON (95% CI 0.248-0.429) and 83% of predictions within +/-0.7 RON; cross-validated MAE on development data was 0.590 RON. **Limits:** a different 785 nm research instrument, only 179 samples, no field conditions, and **not Blaise data**. **Phase II plan:** native Blaise spectra across suppliers, grades, batches, operators and temperatures, each sample with an independent laboratory RON, split by physical sample; then validate or recalibrate the thresholds. Timeline: M1 sampling protocol and data, M2 model on native spectra, M3 app, M4 validation study.

6. TEAM STRENGTH AND READINESS

Solo entrant: Umar Tabbsum. Dual B.S. in computer engineering technology and electronics engineering technology; former EMC engineer and test-lab manager who helped build an ISO 17025/A2LA-accredited lab; experience in test automation (LabVIEW, GPIB) and qualification testing. That measurement-and-validation background shapes this plan: lab-referenced data, held-out testing and honest uncertainty. The research baseline above is already built and reproducible.

7. SUCCESS METRICS

Best Data: native Blaise spectra paired with independent lab RON across multiple suppliers, grades, batches, operators and temperatures, with replicate consistency reported. **Best User Interface:** scan to a plain three-state status in seconds, estimates always labeled as estimates, tested for unaided first-time completion. **Best Impact:** share of off-distribution samples correctly routed to lab confirmation, and agreement with lab RON in the Routine tier. Numeric targets will be set from Phase II data rather than promised now.

8. COMMERCIALIZATION PATHWAY

Release on the Apple App Store and Google Play as a subscription per site or fleet, used with Blaise devices and consumables purchased separately (supply terms to be confirmed with Forward Edge-AI). **Illustrative** path to $1M annual gross revenue: about 330 paying sites at roughly $250 per month, plus per-scan or enterprise tiers. These are planning assumptions not yet tested; Phase II includes interviews with fleet and depot managers to validate pricing and demand.

TEAM AND CONTACT

Team: Solo entry | **Primary contact:** Umar Tabbsum | **Email:** umartabbsum@yahoo.com | **Affiliation:** Independent | **Location:** Dublin, Ohio, USA

Research proof of concept on a public dataset; not native Blaise performance, not field- or regulatory-validated, and not a replacement for standardized laboratory RON testing. Submitting grants Forward Edge-AI the promotional license described in the official pitch template.